

E-Commerce Sales Analysis

Python · Pandas · SQL · Matplotlib | Full Year 2023

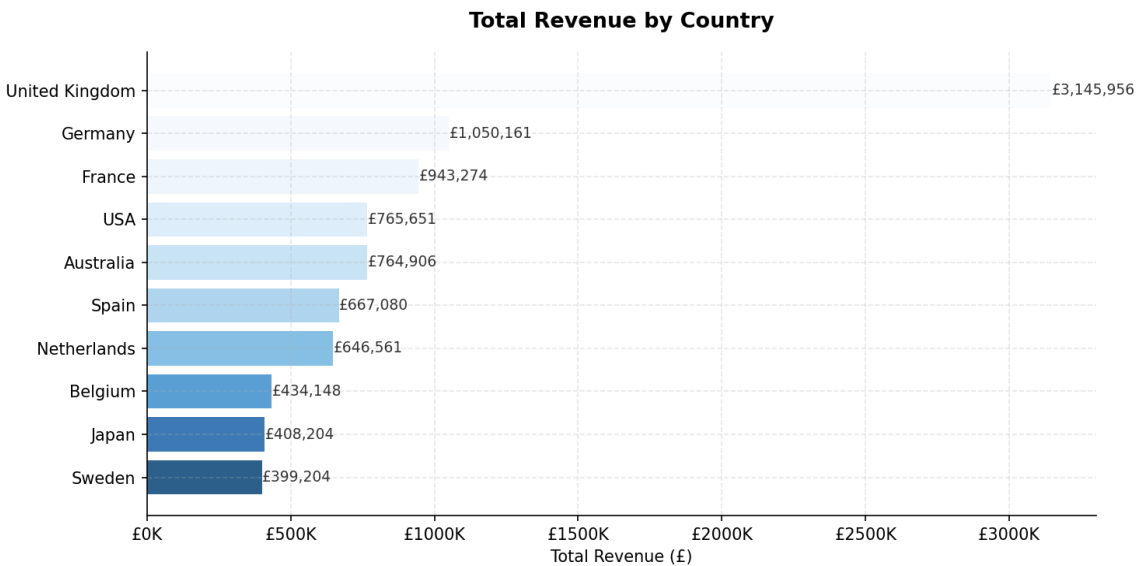
Project Overview

This project analyses 5,000 e-commerce transactions across 10 countries for the full year 2023. The dataset was cleaned using Python and Pandas, removing 178 records with missing customer IDs and zero-value prices. Key business questions were answered through data aggregation, grouping, and visualisation.

Key Metrics

Total Revenue	Total Orders	Total Customers	Avg Order Value
£9,225,145	4,822	400	£1,913

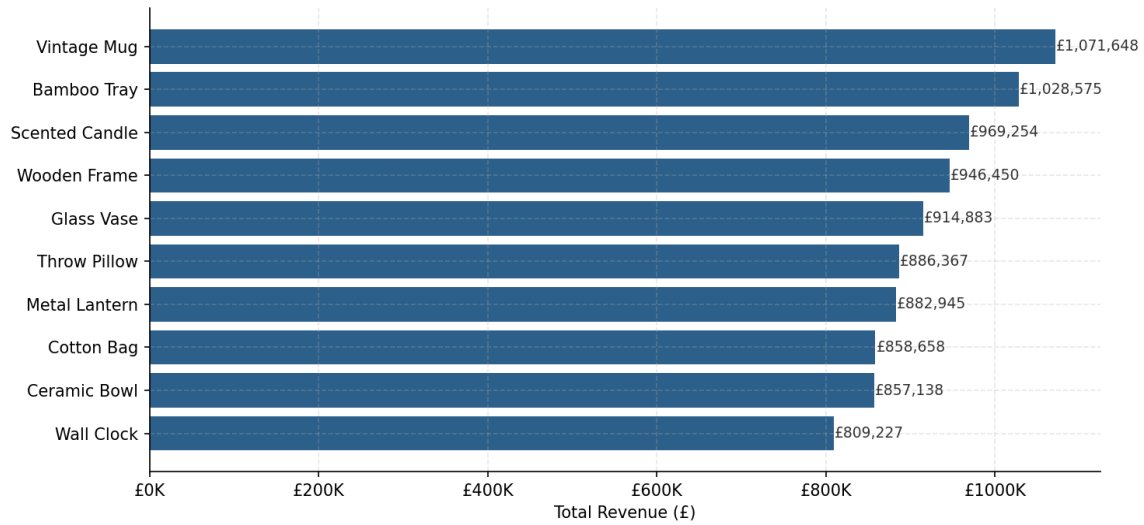
Q1 — Total Revenue by Country



Insight: United Kingdom is the top revenue-generating country at £3.1M, accounting for 34% of total revenue. Germany and France follow at £1.05M and £943K respectively.

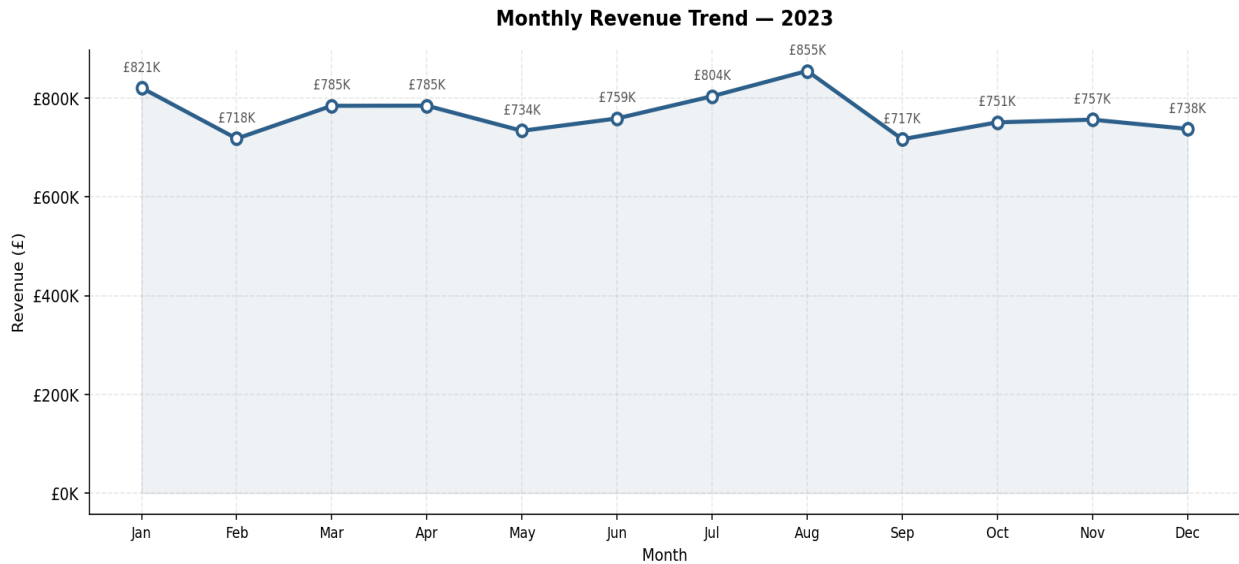
Q2 — Top Products by Revenue

Top Products by Revenue



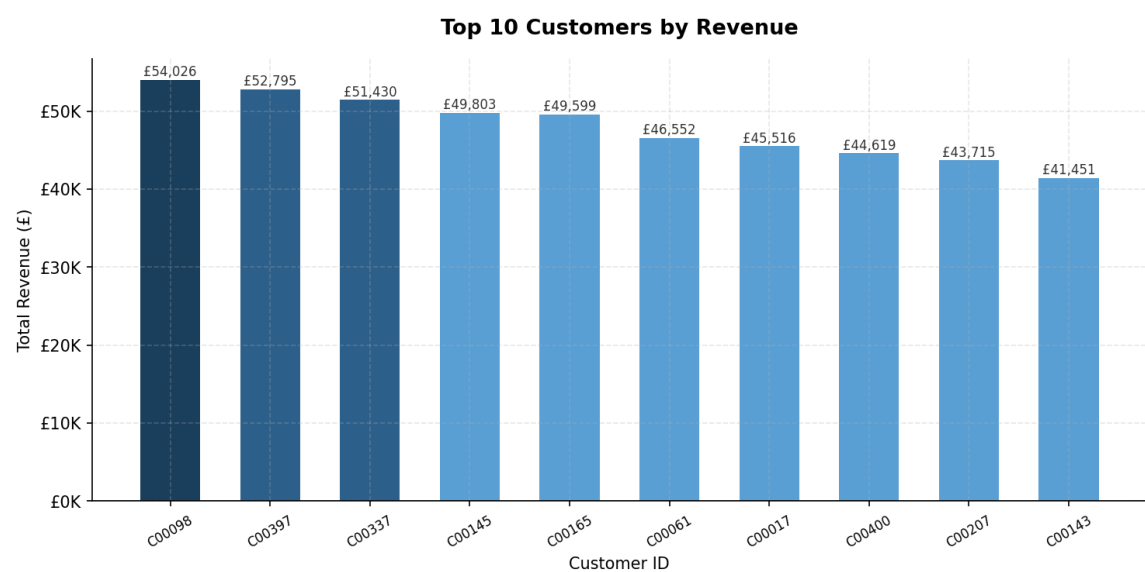
Insight: Vintage Mug is the best-selling product at £1.07M, closely followed by Bamboo Tray at £1.03M. Revenue is relatively evenly spread across all 10 product categories.

Q3 — Monthly Revenue Trend



Insight: August recorded the highest monthly revenue at £855K. Revenue dipped slightly in February and September. Overall trend remained stable throughout 2023 with no major seasonal spike.

Q4 — Top 10 Customers by Revenue



Insight: Top customer C00098 generated £54,026 alone. The top 10 customers together contributed approximately £479K, representing 5.2% of total revenue — indicating a relatively healthy, diversified customer base.

Revenue Breakdown by Country

Country	Total Revenue	Share %
United Kingdom	£3,145,956	34.1%
Germany	£1,050,161	11.4%
France	£943,274	10.2%
USA	£765,651	8.3%
Australia	£764,906	8.3%
Spain	£667,080	7.2%
Netherlands	£646,561	7.0%
Belgium	£434,148	4.7%
Japan	£408,204	4.4%
Sweden	£399,204	4.3%

SQL Queries Used

The following SQL queries were written to extract business insights from the dataset:

```
-- Q1: Total Revenue by Country
SELECT Country, ROUND(SUM(Quantity * UnitPrice), 2) AS Total_Revenue
FROM ecommerce
WHERE UnitPrice > 0 AND Quantity > 0 AND CustomerID IS NOT NULL
GROUP BY Country
```

```
ORDER BY Total_Revenue DESC;

-- Q2: Top Products by Revenue
SELECT Description, ROUND(SUM(Quantity * UnitPrice), 2) AS Revenue
FROM ecommerce
GROUP BY Description
ORDER BY Revenue DESC
LIMIT 10;

-- Q3: Monthly Revenue Trend
SELECT MONTH(InvoiceDate) AS Month,
       ROUND(SUM(Quantity * UnitPrice), 2) AS Monthly_Revenue
FROM ecommerce
GROUP BY MONTH(InvoiceDate)
ORDER BY Month;

-- Q4: Top 10 Customers
SELECT CustomerID, ROUND(SUM(Quantity * UnitPrice), 2) AS Total_Spent
FROM ecommerce
WHERE CustomerID IS NOT NULL
GROUP BY CustomerID
ORDER BY Total_Spent DESC
LIMIT 10;

-- Q5: Average Order Value
SELECT ROUND(AVG(order_total), 2) AS Avg_Order_Value
FROM (
    SELECT InvoiceNo, SUM(Quantity * UnitPrice) AS order_total
    FROM ecommerce GROUP BY InvoiceNo
) t;
```

Conclusions & Recommendations

1. The United Kingdom is by far the largest market. Investing more in UK marketing would yield the highest return.
2. Vintage Mug and Bamboo Tray are the top products. These should be kept well stocked and promoted.
3. Revenue is stable across all months with no sharp seasonal drop — indicating a reliable, consistent customer base.
4. The top 10 customers contribute only 5% of revenue, which means the business is not over-reliant on any single customer — a healthy sign.
5. A loyalty programme targeting the top 50 customers (who likely contribute 15–20% of revenue) could further increase retention and lifetime value.